TRIAGE-LINK

Offline-first PWA for field casualty documentation & coordination

Capability overview — the PWA, the encryption & audit layer, the hosted multi-tenant backend with admin security, and the market flavors (Humanitarian / NGO · Ontario EMS / regulated · productized backend).

React + TypeScript · IndexedDB + op-log · WebCrypto · Fastify · NEMSIS/OADS

POSITIONING

One field record, no signal required

- Document a casualty completely OFFLINE — injuries, vitals, treatments, photos, triage, handover — with no server in the loop.
- Built for where connectivity is unreliable or absent: disaster/MCI, humanitarian field clinics, mass-gathering and prehospital EMS.
- Installs as a PWA; data lives locally in IndexedDB and survives reloads, crashes, and power cycles.
- Optional, never-required sync aggregates records across a team only where a deployment wants it.
- One codebase, three market 'flavors' that share the offline-first core.

ARCHITECTURE

Offline-first core, optional everything else

DEVICE — PWA

- React + TS UI
- IndexedDB (Dexie)
- Op-log sync engine
- WebCrypto vault
- Operators + audit

→

OPTIONAL — Sync service

- Fastify, multi-tenant
- Conflict-aware /sync
- OIDC admin + console
- Quota · retention · limits
- PostgreSQL

→

OPTIONAL — EHR gateway

- ONE ID / Ontario Health
- PCR $match · handover
- mTLS, server-side
- NEMSIS/OADS export
- Audit trail

The PWA is fully functional with neither optional tier present. Sync and the EHR gateway are additive, server-side, and off by default.

PWA · CAPTURE

Documenting a casualty

Body & injuries

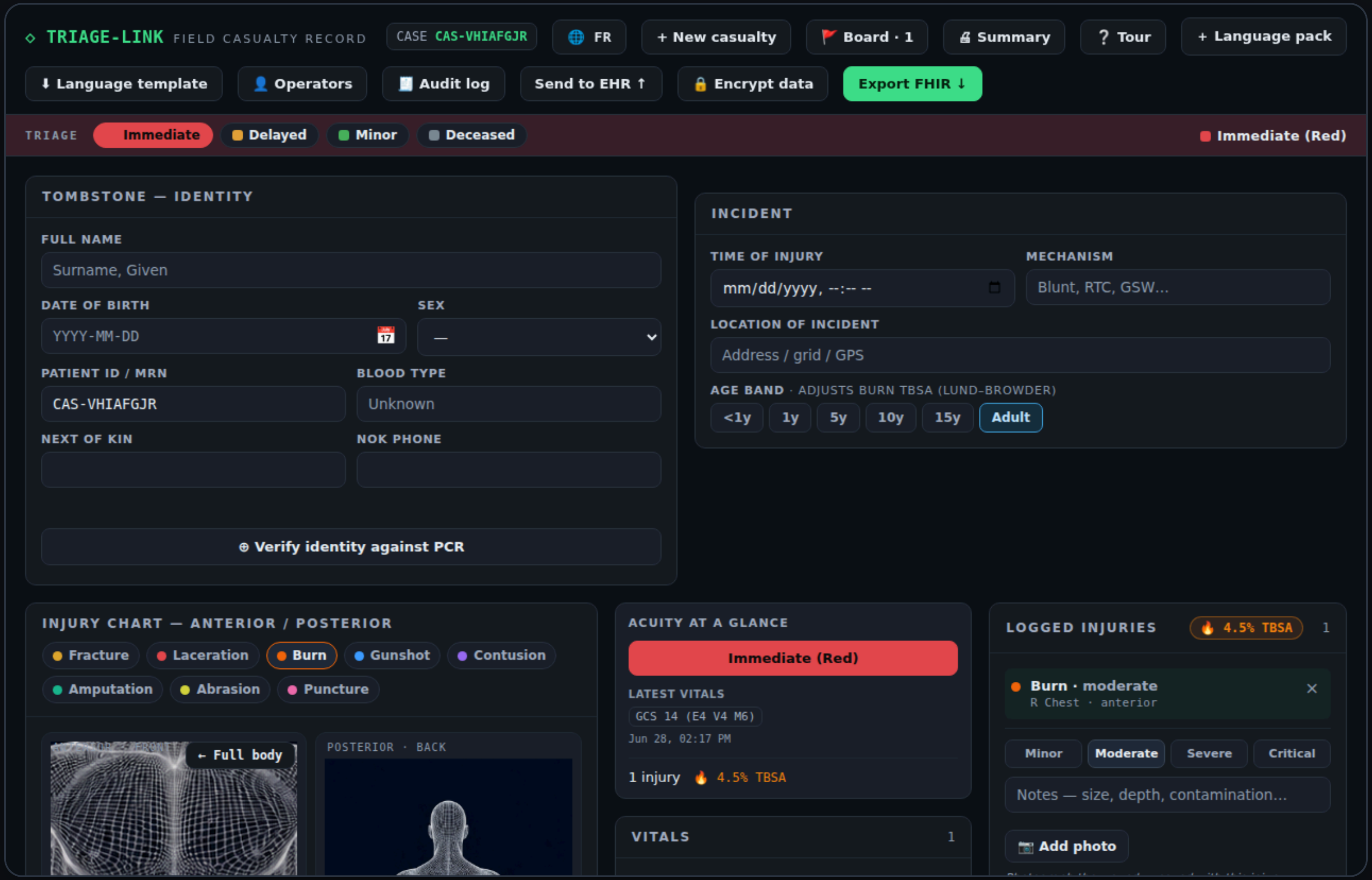
- Tap-to-mark injuries on an anterior/posterior body chart
- Injury palette (GSW, burn, laceration, fracture…) + severity
- Per-injury notes + wound photos (stored as out-of-line blobs)
- Burn TBSA auto-estimate (Lund–Browder by age band)

Triage & identity

- START-style tag: Immediate / Delayed / Minor / Deceased
- Patient: name, DOB→age band, sex, MRN, NOK, blood type
- Incident: time, mechanism, location
- Color-coded multi-casualty Triage Board

PWA · SCREENSHOT

The field record — one screen, fully offline



Triage tag · patient identity · incident · injury body-chart · acuity glance (GCS, TBSA) · vitals — captured on-device.

Vitals, interventions & trends

Vitals & scoring

- Timestamped vital sets: HR, BP, RR, SpO_2, GCS, pain
- Built-in GCS calculator (eye/verbal/motor → total)
- Vitals-trend sparklines once ≥2 readings exist
- Time-since-injury clock (T+) on the record

Treatments

- Structured log: tourniquet, airway, decompression, IV/fluids,
- medication, splinting, wound packing, CPR…
- Each entry timestamped + attributed to the on-duty operator
- Feeds the AT-MIST handover summary

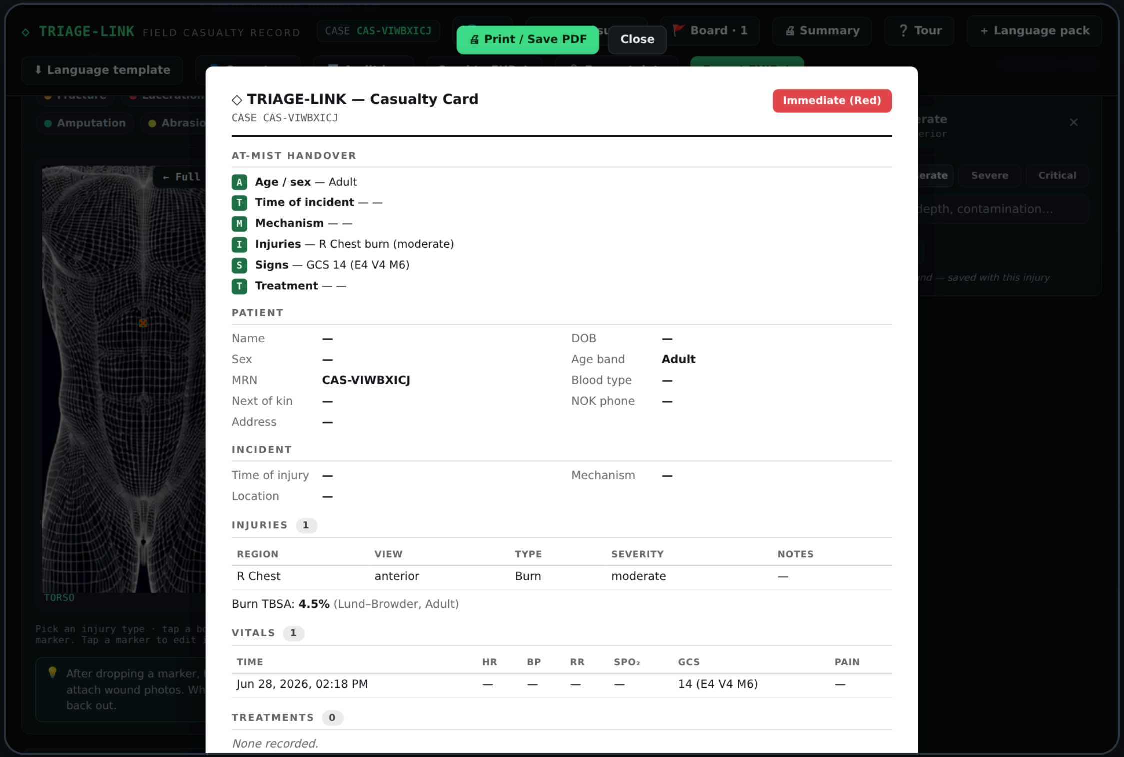
PWA · HANDOVER

Summaries & clean handoff

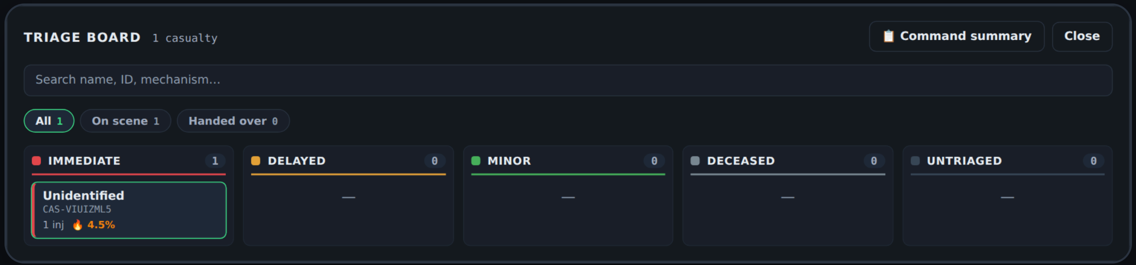
- One-page printable Casualty Summary card (AT-MIST) — print or save as PDF for handover.
- Scene Summary / command roll-up: casualties tallied by triage, on-scene vs handed-over.
- Handover sign-off (who took over care, facility, time) emitted as a FHIR handover bundle.
- Optional 'Send to EHR' contributes the handover to a provincial EHR via the gateway.
- Everything renders offline; nothing leaves the device unless you export or sync.

PWA · SCREENSHOT

Handover card & the scene picture



One-page AT-MIST casualty card — print or save as PDF.



Triage Board — every casualty by acuity, on-scene vs handed-over.

PWA · ACCESSIBILITY

Four languages, guided & spoken

Internationalization

- EN / FR / AR / FA built in — Arabic & Persian fully RTL
- Loadable JSON language packs: add a language with NO app release
- Downloadable English template to translate; parity-tested in CI
- Natural wording, not literal — reviewed per language

Guided tour

- Smart tour highlights each real control
- Offline voice-over (SpeechSynthesis) in the active language
- Action steps auto-advance once the user does them
- Every user-visible feature is taught in the tour (enforced)

PWA · PLATFORM

Installable, offline, durable

- Installable PWA (service worker + Workbox precache) — launches and runs with no network.
- All data in IndexedDB via Dexie; records, op-log, photos, audit chain persist locally.
- Phones, tablets, laptops; responsive layout collapses gracefully on small screens.
- No telemetry, no implicit network calls — a casualty is documented entirely on-device.

DATA INTEGRITY

Conflict-aware op-log sync engine

- Every change is journaled as an immutable operation (scalars + collections), not a blind overwrite.
- Deterministic resolve(): Lamport clocks order edits; ties break predictably — same inputs, same result, every device.
- Concurrent edits to different fields all survive; same-field edits pick a deterministic winner and REPORT the conflict (losing op retained).
- The server stores and folds ops — it does not implement its own divergent merge logic.
- Incremental sync (cursor) pulls only what changed; full-state pulls are paginated.

SECURITY · AT REST

Opt-in encryption vault

How it works

- AES-256-GCM, key derived from a passphrase via PBKDF2 (210k iters)
- Encrypts the heaviest PHI — wound photos — plus records & op-log
- Key lives only in memory while unlocked; locking drops it
- Idle auto-lock; wrong passphrase rejected via a verifier

Safety posture

- DEFAULT-OFF: with no vault, behavior is byte-for-byte unchanged
- Mixed plaintext/encrypted rows read correctly through a toggle
- Sealed records unreadable (get/list skip them) while locked
- Crash-safe enable/disable — data is never orphaned

SECURITY · ACCESS

Operators, RBAC-lite & tamper-evident audit

Shared-device access

- Local operator roster (field / lead / admin)
- Records & audit entries attributed to the on-duty operator
- Step-up PIN re-auth gates sensitive actions (delete, export…)
- Empty roster = open (community default); adding operators opts in

Audit log

- Append-only, hash-chained entries (SHA-256 per entry)
- Tampering/deletion breaks the chain — detectable offline
- No update/delete API; reviewable even while the vault is locked
- Covers create / view / delete / export / vault / step-up

DATA PORTABILITY

Backup, restore & export

Backup / restore

- Full JSON backup of every record (plain or encrypted)
- Restore by merge (keep newer of duplicates) or replace
- Encrypted backup keeps PHI unreadable without the passphrase

CSV interchange

- Roster CSV export (identity + incident) for analytics/QA
- CSV import onboards a patient list from paper/another system
- Date-range filter scopes an export to a window
- Deployment provenance stamped on rows (humanitarian)

HOSTED BACKEND

Multi-tenant sync service (Fastify)

- Optional cloud or self-hosted backend for cross-team aggregation — the PWA never requires it.
- Per-tenant isolation: each API key authenticates AND scopes a tenant's data; conflict-aware /sync stores and folds ops.
- Hardened: sanitized error envelopes, paginated pulls, per-tenant storage quota, audit-log retention TTL.
- OpenAPI 3 + Swagger UI; liveness/readiness probes; per-tenant metrics; request-id correlation.
- Runs on PostgreSQL; graceful shutdown; rate-limited per IP.

Admin security & console

Admin authentication

- Tenant-admin API (/admin/*) behind a static token OR OIDC SSO
- OIDC: IdP-issued JWT, audience-checked, optional role mapping
- Provision tenants; issue / rotate / revoke per-tenant API keys
- Every admin mutation written to a separate admin-audit trail

Graphical console

- Opt-in static admin console at /console
- Token-entry auth; holds no secrets (the API gate enforces)
- Browse tenants, keys, metrics & audit from a browser
- Off by default — enabled per deployment

THREE MARKET FLAVORS

One core, productized three ways

Humanitarian / NGO · Ontario EMS (regulated) · Productized backend

FLAVOR · HUMANITARIAN / NGO

Field documentation where the cloud isn't

Shipped on this line

- Deployment context — device-wide operation tag + provenance banner
- Disaster/MCI mode — one toggle makes encryption mandatory + command roll-up
- Kiosk defaults — assign-operator prompt + 2-min auto-lock on shared devices
- Donor export — provenance-stamped CSV + date-range filter

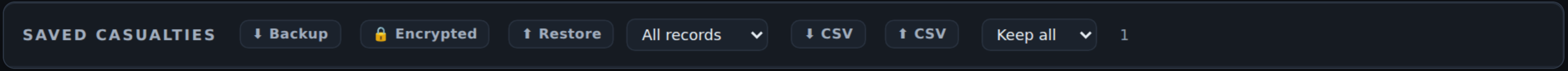
More

- Retention presets — confirmed, step-up-gated purge window (off/30/90/180/365 d)
- Air-gapped packaging — one Docker stack (PWA + sync + Postgres) offline on a laptop
- Loadable language packs for any region; encrypted backups for low-infra handoff
- Intended use = documentation/coordination tool (light reg load)

FLAVOR · HUMANITARIAN — SCREENSHOTS

Built for the multi-team deployment

Deployment context — tag the operation, response type & organization (MCI mode for shared devices).



Donor export — date-range CSV + provenance, plus a retention window (here: 'Keep all').

FLAVOR · ONTARIO EMS (REGULATED)

Toward a conformant ePCR

NEMSIS / OADS conformance

- Section exporter maps captured data → NEMSIS v3.5 / OADS v4.0 shapes
- Deterministic offline XML serializer (shaped, gap-annotated)
- Pluggable validator (cardinality/datatype/value-set) + placeholder ruleset
- Capture panels: eResponse / eTimes / eCrew / eScene — gaps clear at runtime

In-app & integration

- Read-only Conformance view: live gaps + validator issues, in 4 languages
- Clearly labelled NOT certification (placeholder ruleset, rulesetSource shown)
- ONE ID / Ontario Health gateway scaffold (mTLS, server-side) for PCR $match
- Gated next: official-dictionary reconciliation + live ONE ID credentials

FLAVOR · PRODUCTIZED BACKEND

The shared, hardened service line

- The multi-tenant sync service + admin security is the common spine both market branches reuse.
- Multi-tenant isolation, OIDC SSO + role-based admin, per-tenant rate limits & quota, incremental sync.
- EHR-access and admin audit trails — necessary for any regulated, multi-service deployment.
- Cascaded to every flavor branch, so a hardening fix lands everywhere.

STATUS

Where things stand

Done & merged

- Full offline PWA: capture, triage, vitals, handover, summaries
- Encryption vault · operators · step-up · tamper-evident audit
- Multi-tenant backend hardening + graphical admin console
- Humanitarian: all 5 roadmap items (context→MCI→export→retention→air-gap)
- Ontario: NEMSIS pipeline + capture panels + in-app conformance view

Externally gated next

- Ontario PR-2c — official NEMSIS/OADS dictionary + true XSD validation
- Ontario PR-4 — live ONE ID credentials + real mTLS client cert
- SOC 2 Type II / QMS / SaMD evidence (if intended use crosses the device line)
- CAD / hospital-EHR handover partnerships

Document anywhere.

Offline-first by design · encrypted & audited · ready for the cloud only when you want it.

TRIAGE-LINK — PWA · multi-tenant backend · Humanitarian & Ontario EMS flavors